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United States Patent Application Publication

20250286336

Kind Code

A1

Publication Date

September 11, 2025

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CONNECTOR AND CONNECTOR MANUFACTURING METHOD

Abstract

A connector includes a wire including a core wire and an insulation coating covering an outer periphery of the core wire, a terminal to be connected to the core wire, an inner housing including a terminal accommodating portion for accommodating the terminal inside, a rear holder to be attached to a rear part of the inner housing and a collective sealing portion to be mounted into the rear holder. The rear holder includes a wire insertion hole, through which the wire is inserted. The collective sealing portion includes an inner sealing portion for sealing the wire inserted through the wire insertion hole. The insulation coating of the wire includes a sheath portion and a body portion separate from each other. The sheath portion is disposed forward of the body portion and slidable with respect to the core wire.

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Family ID: 1000008669086

Appl. No.: 18/862552

Filed (or PCT Filed): April 24, 2023

PCT No.: PCT/JP2023/016075

Foreign Application Priority Data

JP 2022-077958

May. 11, 2022

Publication Classification

**Int. Cl.: H01R43/24 (20060101); H01R13/426 (20060101); H01R13/506 (20060101);
H01R13/52 (20060101)**

U.S. Cl.:

**CPC H01R43/24 (20130101); H01R13/426 (20130101); H01R13/506 (20130101);
H01R13/5208 (20130101); H01R13/5202 (20130101)**

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a connector and a connector manufacturing method.

BACKGROUND

[0002] Conventionally, a connector described in Japanese Patent Laid-Open Publication No. 2019-145208 (Patent Document 1 below) is known as a connector for accommodating a terminal-equipped wire, in which a terminal is connected to a core wire exposed from an end of a wire. This connector is provided a terminal including a terminal body and a slide portion, a connector housing including a cavity for accommodating the terminal and a rear holder to be locked to the rear end of the terminal. With the slide portion held at a complete locking position with respect to the terminal body, an upper contact portion of the slide portion is in contact with an upper connection piece of the terminal body from above and presses the upper connection piece downward, and a lower contact portion of the slide portion is in contact with a lower connection piece of the terminal body from below and presses the lower connection piece upward. In this way, the core wire is sandwiched from upper and lower sides by the upper and lower connection pieces and the wire and the terminal are electrically connected.

PRIOR ART DOCUMENT

Patent Document

[0003] Patent Document 1: JP 2019-145208 A

SUMMARY OF THE INVENTION

Problems to be Solved

[0004] The above connector is a non-waterproof connector. If an attempt is made to make this connector waterproof, the connector housing is first used as an inner housing, an outer housing to be disposed on an outer peripheral side of this inner housing is provided, and the inner housing is accommodated to be surrounded by the outer housing and the rear holder. Then, it is thought to mount a one-piece rubber plug into a rear end part of the inner housing to waterproof the wire inserted through a wire insertion hole and mount a rubber ring between the rear holder and the outer housing, thereby waterproofing the inside of the rear holder and the inside of the outer housing.

[0005] However, in the configuration of the above waterproof connector, when the wire is inserted into the wire insertion hole, the core wire exposed in a tip part of the wire may damage a sealing portion of the one-piece rubber plug.

Means to Solve the Problem

[0006] The present disclosure is directed to a connector with a wire including a core wire and an insulation coating covering an outer periphery of the core wire, a terminal to be connected to the core wire, an inner housing including a terminal accommodating portion for accommodating the terminal inside, a rear holder to be attached to a rear part of the inner housing, and a collective sealing portion to be mounted into the rear holder, the rear holder including a wire insertion hole, the wire being inserted through the wire insertion hole, the collective sealing portion including an

inner sealing portion for sealing the wire inserted through the wire insertion hole, the insulation coating of the wire including a sheath portion and a body portion separate from each other, and the sheath portion being disposed forward of the body portion and slidable with respect to the core wire.

Effect of the Invention

[0007] According to the present disclosure, it is possible to suppress the damage of a collective sealing portion in a connector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a perspective view of a connector according to an embodiment.

[0009] FIG. 2 is an exploded perspective view of the connector.

[0010] FIG. 3 is a front view of the connector.

[0011] FIG. 4 is a section along A-A of FIG. 3.

[0012] FIG. 5 is a section along B-B of FIG. 3.

[0013] FIG. 6 is a side view of the connector.

[0014] FIG. 7 is a section along C-C of FIG. 6.

[0015] FIG. 8 is a front view of a rear holder integrally configured with a collective sealing portion.

[0016] FIG. 9 is a section along D-D of FIG. 8.

[0017] FIG. 10 is a front view when only the rear holder is viewed from front.

[0018] FIG. 11 is a perspective view in section along E-E of FIG. 10 when viewed obliquely from a front upper side.

[0019] FIG. 12 is a side view when only the rear holder is viewed laterally.

[0020] FIG. 13 is a section along F-F of FIG. 12.

[0021] FIG. 14 is a view in section along A-A of FIG. 3 showing a state where a wire is inserted through a wire insertion hole and a sheath portion is in contact with an inner sealing portion.

[0022] FIG. 15 is a view in section along A-A of FIG. 3 showing a state where the wire is pushed further forward from the state of FIG. 14 and the sheath portion is slid with respect to the core wire.

[0023] FIG. 16 is a view in section along A-A of FIG. 3 showing a state where the wire is pushed further forward from the state of FIG. 15, the core wire is inside the terminal and a body portion is sealed by the inner sealing portion.

DETAILED DESCRIPTION TO EXECUTE THE INVENTION

[Description of Embodiments of Present Disclosure]

[0024] First, embodiments of the present disclosure are listed and described below.

[0025] (1) The connector of the present disclosure is provided with a wire including a core wire and an insulation coating covering an outer periphery of the core wire, a terminal to be connected to the core wire, an inner housing including a terminal accommodating portion for accommodating the terminal inside, a rear holder to be attached to a rear part of the inner housing, and a collective sealing portion to be mounted into the rear holder, the rear holder including a wire insertion hole, the wire being inserted through the wire insertion hole, the collective sealing portion including an inner sealing portion for sealing the wire inserted through the wire insertion hole, the insulation coating of the wire including a sheath portion and a body portion separate from each other, and the sheath portion being disposed forward of the body portion and slidable with respect to the core wire.

[0026] According to this configuration, the front end part of the core wire can be suppressed from contacting the inner sealing portion of the collective sealing portion by inserting the wire into the wire insertion hole with the front end part of the core wire protected by the sheath portion.

Therefore, the damage of the collective sealing portion can be suppressed.

[0027] (2) Preferably, the inner sealing portion seals the body portion.

[0028] According to this configuration, waterproofness by the collective sealing portion can be ensured.

[0029] (3) Preferably, the connector is further provided with an outer housing to be assembled with the rear holder while accommodating the inner housing inside, and an outer sealing portion for sealing between the rear holder and the outer housing.

[0030] According to this configuration, the insides of the inner housing and the rear holder can be waterproofed by providing the outer housing and the outer sealing portion.

[0031] (4) Preferably, the outer sealing portion is integrally formed to the collective sealing portion.

[0032] According to this configuration, the number of components can be reduced since the outer sealing portion is integrally provided to the collective sealing portion.

[0033] (5) Preferably, the collective sealing portion and the rear holder are constituted by one member.

[0034] According to this configuration, mounting man-hours can be reduced since the collective sealing portion needs not be mounted into the rear holder. For example, the collective sealing portion and the rear holder may be configured into one member by forming the collective sealing portion in the rear holder by two-color molding.

[0035] (6) A connector manufacturing method of the present disclosure is a manufacturing method for a connector with a wire including a core wire and an insulation coating covering an outer periphery of the core wire, a terminal, an inner housing including a terminal accommodating portion, a rear holder including a wire insertion hole, and a collective sealing portion to be mounted into the rear holder, the collective sealing portion sealing the wire inserted through the wire insertion hole, and includes accommodating the terminal inside the terminal accommodating portion, attaching the rear holder having the collective sealing portion mounted therein to a rear part of the inner housing, forming a sheath portion and a body portion disposed rearward of the sheath portion by cutting off a front end part of the insulation coating of the wire, protecting a front end part of the core wire by the sheath portion by sliding the sheath portion forward with respect to the core wire, inserting the wire having the front end part of the core wire protected by the sheath portion into the wire insertion hole of the rear holder and bringing the front end part of the core wire to inside of the terminal, and connecting the core wire and the terminal.

[0036] Since such a connector manufacturing method includes forming the sheath portion in the insulation coating and protecting the front end part of the core wire by the sheath portion, the front end part of the core wire can be suppressed from contacting the collective sealing portion when the wire is inserted into the wire insertion hole. Therefore, the damage of the collective sealing portion can be suppressed.

[Details of Embodiment of Present Disclosure]

[0037] Hereinafter, an embodiment of the present disclosure is described. The present disclosure is not limited to these illustrations, but is represented by claims and intended to include all changes in the scope of claims and in the meaning and scope of equivalents.

EMBODIMENT

[0038] The embodiment of the present disclosure is described with reference to FIGS. **1** to **16**. In the following description, a direction indicated by an arrow Z is referred to as an upward direction, a direction indicated by an arrow X is referred to as a forward direction and a direction indicated by an arrow Y is referred to as a leftward direction. Note that, for a plurality of identical members, only some members may be denoted by a reference sign and the other members may not be denoted by the reference sign.

[Connector]

[0039] As shown in FIGS. **1** and **2**, the connector **10** is provided with an inner housing **20**, an outer housing **30**, a sealing ring **35**, terminals **50**, wires **40**, a rear holder **60** and a collective sealing portion **70**.

[Wire]

[0040] The wire **40** is provided with a core wire **41** and an insulation coating **42** made of insulating synthetic resin for surrounding the outer peripheral surface of the core wire **41**. The insulation coating **42** is stripped in an end part of the wire **40** to expose the core wire **41**. The core wire **41** is a so-called single-core wire composed of one metal wire. Note that the core wire **41** may be a stranded wire formed by twisting a plurality of thin metal wires. An arbitrary metal such as copper, copper alloy, aluminum or aluminum alloy can be appropriately selected as a metal constituting the core wire **41** if necessary. The core wire **41** of this embodiment is made of copper or copper alloy.

[Sheath Portion, Body Portion]

[0041] The insulation coating **42** is provided with a sheath portion **43** and a body portion **44** separate from the sheath portion **43** and disposed behind the sheath portion **43**. The sheath portion **43** is slidable in an extension direction (front-rear direction) of the wire **40** with respect to the core wire **41**. The sheath portion **43** is formed by being cut off from the body portion **44** by a wire stripper or the like.

[Terminal]

[0042] The terminal **50** is a female terminal and provided with a terminal body **51** made of metal and a slide portion **52** relatively slidable with respect to the terminal body **51**. The terminal body **51** is formed into a predetermined shape by a known method such as press-working, cutting or casting. An arbitrary metal such as copper, copper alloy, aluminum, aluminum alloy or stainless steel can be appropriately selected as a metal constituting the terminal body **51** if necessary. The terminal body **51** of this embodiment is made of copper or copper alloy. A plating layer may be formed on the surface of the terminal body **51**. An arbitrary metal such as tin, nickel or silver can be appropriately selected as a metal constituting the plating layer if necessary. Tin plating is applied to the terminal body **51** of this embodiment.

[Terminal Body]

[0043] As shown in FIG. **4**, the terminal body **51** includes a connecting tube portion **51A** in the form of a rectangular tube, into which a mating male terminal (not shown) is inserted, a base portion **51B** in the form of a rectangular tube disposed behind the connecting tube portion **51A** and an upper connection piece **51C** and a lower connection piece **51D** extending rearward from the base portion **51B**. The front end of the connecting tube portion **51A** is open to enable the insertion of the mating male terminal.

[0044] A resiliently deformable resilient contact piece **51E** is disposed on a side wall of the connecting tube portion **51A**. The male terminal inserted into the connecting tube portion **51A** contacts the resilient contact piece **51E**. Further, a front end part of the upper wall of the connecting tube portion **51A** is provided with a locking lance **53** made of metal and formed by cutting and raising a part of the connecting tube portion **51A**.

[0045] The upper connection piece **51C** is formed to be resiliently deformable in a vertical direction with a rear end part of the upper wall of the base portion **51B** as a fulcrum. The lower connection piece **51D** is formed to be resiliently deformable in the vertical direction with a rear end part of the lower wall of the base portion **51B** as a fulcrum. The upper and lower connection pieces **51C**, **51D** are formed to have substantially the same length in a front-rear direction.

[0046] The terminal body **51** further includes a terminal window portion **51F** open upward at a position behind the connecting tube portion **51A** and in front of the base portion **51B**. A front end part of the core wire **41** can be detected through the terminal window portion **51F** with the core wire **41** disposed in a space between the upper and lower connection pieces **51C**, **51D**. Detectable means visually confirmable from outside by a worker, detectable from outside by a camera (not shown) or capable of electrically detecting the front end part of the core wire **41** by inserting a probe (not shown) from outside.

[Slide Portion]

[0047] The slide portion **52** is in the form of a rectangular tube extending in the front-rear direction.

The slide portion **52** is formed by a known method such as cutting, casting or press-working if necessary. An arbitrary metal such as copper, copper alloy, aluminum, aluminum alloy or stainless steel can be appropriately selected as a metal constituting the slide portion **52** if necessary. The slide portion **52** of this embodiment is made of copper or copper alloy. A plating layer may be formed on the surface of the slide portion **52**. An arbitrary metal such as tin, nickel or silver can be appropriately selected as a metal constituting the plating layer if necessary. Tin plating is applied to the slide portion **52** of this embodiment.

[0048] A cross-sectional shape of the slide portion **52** is the same as or somewhat larger than that of a region of the terminal body **51** where the upper and lower connection pieces **51C**, **52D** are provided. In this way, the slide portion **52** is externally fittable to the region of the terminal body **51** where the upper and lower connection pieces **51C**, **52D** are provided.

[0049] An upper contact portion **52A** projecting downward is provided on the lower surface of the upper wall of the slide portion **52**. A lower contact portion **52B** projecting upward is provided on the upper surface of the lower wall of the slide portion **52**.

[0050] The slide portion **52** is slidable between a temporary locking position (position of the slide portion **52** in FIGS. **14** to **16**) and a complete locking position (position of the slide portion **52** in FIG. **4**) set to be forward of the temporary locking position in a state externally fit to the region of the terminal body **51** where the upper and lower connection pieces **51C**, **51D** are provided. With the slide portion **52** held at the complete locking position with respect to the terminal body **51**, the upper contact portion **52A** is in contact with the upper surface of the upper connection piece **51C** from above and the lower contact portion **52B** is in contact with the lower surface of the lower connection piece **51D** from below.

[0051] With the slide portion **52** held at the complete locking position with respect to the terminal body **51**, the upper contact portion **52A** presses the upper connection piece **51C** from above, whereby the upper connection piece **51C** is resiliently deformed downward. Further, the lower contact portion **52B** presses the lower connection piece **51D** from below, whereby the lower connection piece **51D** is resiliently deformed upward. In this way, the core wire **41** is sandwiched in the vertical direction by the resiliently deformed upper and lower connection pieces **51C**, **51D** in a state where the core wire **41** is disposed to extend in the front-rear direction in the space between the upper and lower connection pieces **51C**, **51D** and the slide portion **52** is held at the complete locking position with respect to the terminal body **51**. That is, the upper connection piece **51C** contacts the core wire **41** from above by being pressed downward by the upper contact portion **52A**, and the lower connection piece **51D** contacts the core wire **41** from below by being pressed upward by the lower contact portion **52B**.

[0052] A front part of the upper connection piece **51C** projects downward, a rear part of the lower connection piece **51D** projects upward, and these projecting parts are shifted in the front-rear direction. Thus, the core wire **41** is held in a state bent in the vertical direction. In this way, the core wire **41** is electrically connected to the terminal **50** and firmly held not to come out against the rearward pulling of the wire **40**.

[0053] As shown in FIG. **16**, the front end part of the slide portion **52** is provided with a jig contact portion **54** projecting upward from the upper wall. An unillustrated jig comes into contact with the jig contact portion **54** from behind and the slide portion **52** is pushed forward by the jig, whereby the slide portion **52** is movable forward.

[Inner Housing]

[0054] As shown in FIG. **2**, the inner housing **20** is made of insulating synthetic resin and includes a terminal accommodating portion **21** for accommodating the plurality of terminals **50**. The inner housing **20** has a substantially rectangular parallelepiped shape flat in the vertical direction and extending in the lateral direction. The terminal accommodating portion **21** includes a plurality of cavities **21A** penetrating therethrough in the front-rear direction. The plurality of cavities **21** are arranged in parallel at intervals in the lateral direction in two upper and lower stages. The cavities

21A formed in the upper stage and the cavities **21** formed in the lower stage are disposed at positions shifted in the lateral direction when viewed from above. Note that the number of the cavities **21A** is arbitrary and the number of the stages in the vertical direction is also arbitrary. [0055] The unillustrated male terminal is insertable into the cavity **21A** from front. As shown in FIG. 4, the terminal **50** is inserted into the cavity **21A** from behind and can be accommodated inside the terminal accommodating portion **21**.

[0056] With the terminal **50** accommodated in the cavity **21A**, a connector window portion **21B** is formed at a position corresponding to the terminal window portion **51F** of the terminal **50** in the terminal accommodating portion **21**. By this connector window portion **21B**, the terminal window portion **51F** communicates with outside. The terminal window portion **51F** of the terminal **50** is detectable through the connector window portion **21B**. In this way, the front end part of the core wire **41** is detectable from outside through the connector window portion **21B** and the terminal window portion **51F**.

[0057] As shown in FIG. 2, the inner housing **20** includes a separation wall **22** partitioning between the cavities **21A** in the upper stage and the cavities **21A** in the lower stage. The separation wall **22** extends rearward from rear end parts of the cavities **21A**. Partition walls **23** extending in the front-rear direction and projecting upward or downward are respectively provided on the upper and lower surfaces of the separation wall **22**. The terminals **50** accommodated in the respective cavities **21A** are electrically insulated from the terminals **50** adjacent in the lateral direction by these partition walls **23**.

[0058] Temporary lock portions **24** projecting outward are provided at positions near rear end parts of both left and right side walls of the inner housing **20**, and complete locking portions **25** are provided at positions forward of these temporary locking portions **24**.

[0059] As shown in FIG. 4, a lance hole **26** open upward is provided on a front end side of the cavity **21A** in the upper stage, and a lance hole **26** open downward is provided on a front end side of the cavity **21A** in the lower stage. The locking lance **53** of the terminal **50** is locked to an inner wall constituting the lance hole **26**, whereby the terminal **50** is held in a state accommodated in the cavity **21A**.

[Rear Holder]

[0060] As shown in FIGS. 4 and 5, the rear holder **60** is made of insulating synthetic resin and provided with a holder base portion **62**, a holder body **61** projecting forward from the holder base portion **62** and a tubular protection wall **63** disposed on an outer peripheral side of the holder body **61** and projecting forward from the holder base portion **62**. The rear holder **60** is formed by injection-molding the insulating synthetic resin.

[0061] The holder body **61** includes a box-shaped holder receptacle **61A** open forward and a seal mounting portion **61B** provided continuously with a rear side of the holder receptacle **61A**. The holder receptacle **61A** is externally fit to a rear part of the inner housing **20**. As shown in FIG. 2, substantially gate-shaped lock receiving portions **64** are provided on front end parts of both left and right side walls of the holder receptacle **61A**. As shown in FIG. 5, the lock receiving portions **64** are resiliently lockable to the temporary lock portions **24** and the complete lock portions **25** of the inner housing **20**.

[0062] By locking the temporary lock portions **24** of the inner housing **20** and the lock receiving portions **64** of the holder body **61**, the rear holder **60** is held at a temporary locking position (see FIGS. 14 to 16) with respect to the inner housing **20**. Further, by locking the complete lock portions **25** of the inner housing **20** and the lock receiving portions **64** of the holder body **61**, the rear holder **60** is held at a complete locking position (position of the rear holder **60** shown in FIG. 5) with respect to the inner housing **20**.

[0063] As shown in FIGS. 8 and 10, the rear holder **60** includes a plurality of wire insertion holes **65**, through which the wires **40** are inserted. The plurality of wire insertion holes **65** are arranged in parallel in the lateral direction and provided in two upper and lower stages. The respective wire

insertion holes **65** are provided at positions corresponding to the respective cavities **21A** of the inner housing **20**. An inner diameter of the wire insertion hole **65** is set to be equal to or somewhat larger than an outer diameter of the insulation coating **42** of the wire **40**.

[0064] As shown in FIG. **9**, a separation wall receiving portion **61C** is provided near a vertical center in the back wall of the holder receptacle **61A**. This separation wall receiving portion **61C** is open forward and a vertical dimension of an opening is set to be equal to or somewhat larger than a vertical thickness of the separation wall **22** of the inner housing **20**.

[0065] As shown in FIG. **14**, with the rear holder **60** held at the temporary locking position with respect to the inner housing **20**, the separation wall receiving portion **61C** of the holder body **61** is located behind the rear end edge of the separation wall **22** of the inner housing **20**. As shown in FIG. **4**, with the rear holder **60** held at the complete locking position with respect to the inner housing **20**, the separation wall **22** of the inner housing **20** is fit in the opening of the separation wall receiving portion **61C** of the rear holder **60**. In this way, a position shift of the rear holder **60** in the vertical direction with respect to the inner housing **20** is suppressed.

[0066] A step **61D** is formed at the front end position of the wire insertion hole **65** on the inner wall of the holder receptacle **61A** as shown in FIG. **9**. This step **61D** comes into contact with a rear end part of the slide portion **52** from behind when the rear holder **60** is moved from the temporary locking position to the complete locking position with respect to the inner housing **20**. In this way, the terminal **50** is held in a state accommodated in the cavity **21A** of the inner housing **20** and prevented from coming out rearward.

[0067] As shown in FIGS. **11** and **13**, the seal mounting portion **61B** is constituted by a pair of upper and lower mounting walls **61E**. The front end of the upper mounting wall **61E** is coupled to the rear end of the upper wall of the holder receptacle **61A**, and the rear end of the upper mounting wall **61E** is coupled to the holder base portion **62**. The front end of the lower mounting wall **61E** is coupled to the rear end of the lower wall of the holder receptacle **61A**, and the rear end of the lower mounting wall **61E** is coupled to the holder base portion **62**. The collective sealing portion **70** is integrally mounted on the seal mounting portion **61B** by two-color molding.

[0068] As shown in FIGS. **10** and **11**, the protection wall **63** has a tubular shape open forward. The protection wall **63** has an oval shape long in the lateral direction when viewed from front. The holder body **61** is accommodated inside the protection wall **63**. A pair of through holes **63A** are provided in both left and right side walls of the protection wall **63**. As shown in FIGS. **6** and **12**, the through hole **63A** has a rectangular shape long in the front-rear direction when viewed laterally.

[0069] As shown in FIGS. **12** and **13**, a space between the pair of upper and lower mounting walls **61E** are aligned with the through holes **63A** when the rear holder **60** is viewed laterally. Therefore, the lower surface of the upper mounting wall **61E** is disposed at the same height as the upper edge of the through hole **63A**, and the upper surface of the lower mounting wall **61E** is disposed at the same height as the lower edge of the through hole **63A**.

[Collective Sealing Portion]

[0070] The collective sealing portion **70** is made of rubber and, as shown in FIG. **7**, provided with an inner sealing portion **71** disposed between the pair of mounting walls **61E** and an outer sealing portion **72** disposed on an outer peripheral side of the inner sealing portion **71**.

[0071] The inner sealing portion **71** seals the wires **40** inserted through the wire insertion holes **65**. As shown in FIG. **9**, the inner sealing portion **71** includes inner peripheral lips **71A** having a smallest inner diameter near a center in the front-rear direction. An inner diameter of the inner peripheral lip **71A** is smaller than the outer diameter of the wire **40**.

[0072] The outer sealing portion **72** includes outer peripheral lips **72A** formed on the outer peripheral surface of the outer sealing portion **72**. The outer peripheral lips **72A** are provided over the entire periphery of the outer sealing portion **72**. The outer peripheral lips **72A** are covered by the protection wall **63**. The outer sealing portion **72** seals between the outer housing **30** and the rear holder **60**.

[Outer Housing]

[0073] As shown in FIG. 4, the outer housing **30** is made of insulating synthetic resin and provided with a housing accommodating portion **31**, a seal accommodating portion **32** disposed behind the housing accommodating portion **31**, a front receptacle **33** disposed on an outer peripheral side of the housing accommodating portion **31** and open forward and a rear receptacle **34** disposed behind the front receptacle **33** and open rearward. As shown in FIG. 3, a plurality of terminal entrance holes **31A** disposed to correspond to the positions of the plurality of cavities **21A** of the inner housing **20** are provided in the front wall of the housing accommodating portion **31**. The terminal entrance hole **31A** is a hole, into which the male terminal (not shown) to be connected to the terminal **50** is inserted.

[0074] The inside of the housing accommodating portion **31** and that of the seal accommodating portion **32** communicate. A dimension in the front-rear direction of the housing accommodating portion **31** is larger than that of the seal accommodating portion **32** and about three times as large as that of the seal accommodating portion **32**. A vertical dimension of the seal accommodating portion **32** is larger than that of the housing accommodating portion **31** as shown in FIG. 4, and a lateral dimension of the seal accommodating portion **32** is larger than that of the housing accommodating portion **31** as shown in FIG. 5.

[0075] The outer sealing portion **72** of the collective sealing portion **70** is accommodated in a sealed state inside the seal accommodating portion **32**. The housing accommodating portion **31** includes a rear housing accommodating portion **31B** for accommodating the holder receptacle **61A**. The rear receptacle **34** is constituted by the rear housing accommodating portion **31B** and the seal accommodating portion **32**.

[0076] As shown in FIG. 5, the rear receptacle **34** includes lock protrusions **34A** for holding a state where the protection wall **63** of the rear holder **60** is mounted along the outer peripheral surface of the rear receptacle **34** (state where the rear receptacle **34** is fit in the protection wall **63**) by being locked to the inner walls of the through holes **63A**. The lock protrusions **34A** are located near the front end of the seal accommodating portion **32**. With the rear receptacle **34** fit in the protection wall **63**, the outer peripheral lips **72A** of the outer sealing portion **72** are held in close contact with the seal accommodating portion **32**, thereby sealing between the outer housing **30** and the rear holder **60**.

[0077] The front receptacle **33** extends forward to the front end of the housing accommodating portion **31** after extending radially outward from the front end of the rear receptacle **34**. The sealing ring **35** is mounted on the outer peripheral surface of the housing accommodating portion **31**. A mating receptacle (not shown) of a mating housing can enter between the outer peripheral surface of the housing accommodating portion **31** and the inner peripheral surface of the front receptacle **33**, and sealing is provided between the mating housing and the outer housing **30** by sandwiching the sealing ring **35** between the inner peripheral surface of the mating receptacle and the outer peripheral surface of the housing accommodating portion **31**.

[Connector Manufacturing Method]

[0078] Next, an example of a manufacturing method of the connector **10** of this embodiment is described. The manufacturing method of the connector **10** is not limited to the one described below.

[0079] The terminal body **51** and the slide portion **52** are formed by a known method. The slide portion **52** is assembled with the terminal body **51** from behind, and held at the temporary locking position. In this way, the terminal **50** is obtained.

[0080] By injection-molding the synthetic resin, the inner housing **20**, the outer housing **30** and the rear holder **60** are formed. The terminals **50** are inserted into the cavities **21A** of the inner housing **20** from behind. The rear holder **60** and the collective sealing portion **70** are integrally formed by two-color molding by injecting liquid rubber as a raw material of the collective sealing portion **70** into the rear holder **60**. In this way, the inner sealing portion **71** and the outer sealing portion **72** are constituted by one member. This leads to a reduction in the number of components and a reduction

in man-hours since a step of mounting the collective sealing portion 70 is eliminated.

[0081] The rear holder 60 is assembled with the rear end part of the inner housing 20 from behind. Then, front end parts of the lock receiving portions 64 of the rear holder 60 come into contact with the temporary lock portions 24 of the inner housing 20 from behind and the lock receiving portions 64 are expanded and deformed. If the rear holder 60 is pushed further forward, the lock receiving portions 64 are restored and the lock receiving portions 64 of the rear holder 60 are resiliently locked to the temporary lock portions 24 of the inner housing 20. In this way, the rear holder 60 is held at the temporary locking position with respect to the inner housing 20. In this state, the holder receptacle 61A of the rear holder 60 is disposed at a position separated rearward from the rear end of the inner housing 20.

[0082] By stripping the insulation coating 42 in the end part of the wire 40, the core wire 41 is exposed over a predetermined length. The front end part of the insulation coating 42 of the wire 40 is cut off by a wire stripper or the like to form the sheath portion 43 and the body portion 44. By sliding the sheath portion 43 forward with respect to the core wire 41, the front end part of the core wire 41 is protected by the sheath portion 43 (see FIG. 2).

[0083] If the wire 40 having the front end part of the core wire 41 protected by the sheath portion 43 is inserted into the wire insertion hole 65 of the rear holder 60, the sheath portion 43 comes into contact with the inner sealing portion 71 as shown in FIG. 14. In this way, since the front end part of the core wire 41 is protected by the sheath portion 43 in this embodiment, the inner sealing portion 71 is hardly damaged due to the contact of the front end part of the core wire 41 with the inner sealing portion 71. If the wire 40 is pushed further forward, the core wire 41 is fed forward while the sheath portion 43 is kept in contact with the inner sealing portion 71 as shown in FIG. 15.

[0084] If the wire 40 is pushed further forward, the core wire 41 is introduced into the inside of the slide portion 52 through the rear end part of the slide portion 52. Then, as shown in FIG. 16, the front end part of the core wire 41 enters the terminal body 51 and reaches the space between the upper and lower connection pieces 51C, 51D. During this time, the sheath portion 43 is pushed out from behind by the body portion 44, thereby being disposed forward of the inner sealing portion 71. The inner peripheral lip 71A is held in close contact with the body portion 44.

[0085] By seeing the front end part of the core wire 41 or detecting this front end part by a probe or the like in the terminal window portion 51F visually confirmable through the connector window portion 21B, it can be confirmed that the core wire 41 has reached a predetermined position.

[0086] Subsequently, if the jig (not shown) is pushed into contact with the jig contact portion 54 from behind and the slide portion 52 is slid forward, the slide portion 52 is held at the complete locking position with respect to the terminal body 51. With the slide portion 52 held at the complete locking position with respect to the terminal body 51, the upper contact portion 52A of the slide portion 52 contacts the upper connection piece 51C of the terminal body 51 from above to press the upper connection piece 51C downward as shown in FIG. 4. Further, the lower contact portion 52B of the slide portion 52 contacts the lower connection piece 51D of the terminal body 51 from below to press the lower connection piece 51D upward. In this way, the core wire 41 is sandwiched from upper and lower sides by the upper and lower connection pieces 51C, 51D and the terminal 50 and the wire 40 are electrically connected.

[0087] Subsequently, the rear holder 60 is pushed further forward with respect to the inner housing 20. Then, the front end parts of the lock receiving portions 64 of the rear holder 60 ride on the complete lock portions 25 and are expanded and deformed. If the rear holder 60 is pushed further forward, the lock receiving portions 64 are restored and the lock receiving portions 64 of the rear holder 60 are resiliently locked to the complete lock portions 25 of the inner housing 20. In this way, the rear holder 60 is held at the complete locking position with respect to the inner housing 20. In this state, the holder receptacle 61A of the rear holder 60 is externally fit to the rear part of the inner housing 20 and the inner housing 20 and the rear holder 60 are integrally configured.

[0088] As shown in FIG. 4, with the rear holder 60 held at the complete locking position with

respect to the inner housing **20**, the inner peripheral lip **71A** is held in close contact with the body portion **44**. In this way, sealability by the inner sealing portion **71** can be ensured. If the inner peripheral lip **71A** is held in close contact with the vicinity of a boundary between the sheath portion **43** and the body portion **44** unlike this embodiment, water may intrude through a gap between the sheath portion **43** and the body portion **44**.

[0089] Subsequently, the rear holder **60** is assembled with the outer housing **30** from behind, whereby the inner housing **20** is accommodated into the housing accommodating portion **31**. The protection wall **63** of the rear holder **60** is expanded and deformed and rides on the lock protrusions **34A** of the outer housing **30**. If the rear holder **60** is pushed further forward, the protection wall **63** is restored and the lock protrusions **34A** are resiliently locked to the inner walls of the through holes **63A**. In this way, the rear holder **60** is held in a state mounted on the outer housing **30**. In this state, the outer sealing portion **72** of the collective sealing portion **70** is accommodated inside the seal accommodating portion **32** and the outer peripheral lips **72A** are held in close contact with the inner peripheral surface of the seal accommodating portion **32**. In the above way, the manufacturing of the connector **10** is completed.

[Functions and Effects of Embodiment]

[0090] According to the embodiment, the following functions and effects are achieved.

[0091] The connector **10** according to the embodiment is provided with the wires **40** each including the core wire **41** and the insulation coating **42** covering the outer periphery of the core wire **41**, the terminals **50** to be connected to the core wires **41**, the inner housing **20** including the terminal accommodating portion **21** for accommodating the terminals **50** inside, the rear holder **60** to be attached to the rear part of the inner housing **20** and the collective sealing portion **70** to be mounted into the rear holder **60**. The rear holder **60** includes the wire insertion holes **65**, through which the wires **40** are inserted. The collective sealing portion **70** includes the inner sealing portion **71** for sealing the wires **40** inserted through the wire insertion holes **65**. The insulation coating **42** of the wire **40** includes the sheath portion **43** and the body portion **44** separate from each other. The sheath portion **43** is disposed forward of the body portion **44** and slidable with respect to the core wire **41**.

[0092] According to this configuration, by inserting the wire **40** into the wire insertion hole **65** with the front end part of the core wire **41** protected by the sheath portion **43**, the front end part of the core wire **41** can be suppressed from contacting the inner sealing portion **71** of the collective sealing portion **70**. Therefore, the damage of the collective sealing portion **70** can be suppressed.

[0093] In the embodiment, the inner sealing portion **71** seals the body portions **44**.

[0094] According to this configuration, waterproofness by the collective sealing portion **70** can be ensured.

[0095] The connector **10** according to the embodiment is further provided with the outer housing **30** to be assembled with the rear holder **60** while accommodating the inner housing **20** inside, and the outer sealing portion **72** for sealing between the rear holder **60** and the outer housing **30**.

[0096] According to this configuration, the insides of the inner housing **20** and the rear holder **60** can be waterproofed by providing the outer housing **30** and the outer sealing portion **72**.

[0097] In the embodiment, the outer sealing portion **72** is integrally formed to the collective sealing portion **70**.

[0098] According to this configuration, since the outer sealing portion **72** is integrally provided to the collective sealing portion **70**, the number of components can be reduced.

[0099] In the embodiment, the collective sealing portion **70** and the rear holder **60** are constituted by one member.

[0100] According to this configuration, since the collective sealing portion **70** needs not be mounted into the rear holder **60**, mounting man-hours can be reduced. For example, the collective sealing portion **70** and the rear holder **60** may be configured into one member by forming the collective sealing portion **70** in the rear holder **60** by two-color molding.

[0101] The manufacturing method of the connector **10** according to the embodiment is a manufacturing method for the connector **10** provided with the wires **40** each including the core wire **41** and the insulation coating **42** covering the outer periphery of the core wire **41**, the terminals **50**, the inner housing **20** including the terminal accommodating portion **21**, the rear holder **60** including the wire insertion holes **65** and the collective sealing portion **70** to be mounted into the rear holder **60** for sealing the wires **40** inserted through the wire insertion holes **65**, and includes accommodating the terminals **50** inside the terminal accommodating portion **21**, attaching the rear holder **60** having the collective sealing portion **70** mounted therein to the rear part of the inner housing **20**, cutting off the front end part of the insulation coating **42** of the wire **40** to form the sheath portion **43** and the body portion **44** disposed behind the sheath portion **43**, protecting the front end part of the core wire **41** by the sheath portion **43** by sliding the sheath portion **43** forward with respect to the core wire **41**, inserting the wire **40** having the front end part of the core wire **41** protected by the sheath portion **43** into the wire insertion hole **65** of the rear holder **60** and bringing the front end part of the core wire **41** to the inside of the terminal **50**, and connecting the core wire **41** and the terminal **50**.

[0102] Since such a manufacturing method of the connector **10** includes forming the sheath portion **43** in the insulation coating **42** and protecting the front end part of the core wire **41** by the sheath portion **43**, the front end part of the core wire **41** can be suppressed from contacting the collective sealing portion **70** when the wire **40** is inserted into the wire insertion hole **65**. Therefore, the damage of the collective sealing portion **70** can be suppressed.

OTHER EMBODIMENTS

[0103] (1) Although the outer sealing portion **72** and the collective sealing portion **70** are integrally formed in the above embodiment, there is no limitation to this. An outer sealing portion may be separate from a collective sealing portion.

[0104] (2) Although the collective sealing portion **70** and the rear holder **60** are configured as one member in the above embodiment, there is no limitation to this and a collective sealing portion and a rear holder may be configured as separate members.

LIST OF REFERENCE NUMERALS

[0105] **10**: connector [0106] **20** inner housing [0107] **21**: terminal accommodating portion [0108] **21A**: cavity [0109] **21B**: connector window portion [0110] **22**: separation wall [0111] **23**: partition wall [0112] **24**: temporary lock portion [0113] **25** complete lock portion [0114] **26** lance hole [0115] **30**: outer housing [0116] **31**: housing accommodating portion [0117] **31A**: terminal entrance hole [0118] **31B**: rear housing accommodating portion [0119] **32** seal accommodating portion [0120] **33**: front receptacle [0121] **34**: rear receptacle [0122] **34A**: lock protrusion [0123] **35**: sealing ring [0124] **40**: wire [0125] **41**: core wire [0126] **42**: insulation coating [0127] **43**: sheath portion [0128] **44**: body portion [0129] **50**: terminal [0130] **51**: terminal body [0131] **51A**: connecting tube portion [0132] **51B**: base portion [0133] **51C**: upper connection piece [0134] **51D**: lower connection piece [0135] **51E**: resilient contact piece [0136] **51F**: terminal window portion [0137] **52**: slide portion [0138] **52A**: upper contact portion [0139] **52B**: lower contact portion [0140] **53**: locking lance [0141] **54**: jig contact portion [0142] **60** rear holder [0143] **61**: holder body [0144] **61A**: holder receptacle [0145] **61B**: seal mounting portion [0146] **61C**: separation wall receiving portion [0147] **61D**: step [0148] **61E**: mounting wall [0149] **62**: holder base portion [0150] **63**: protection wall [0151] **63A**: through hole [0152] **64**: lock receiving portion [0153] **65** wire insertion hole [0154] **70**: collective sealing portion [0155] **71** inner sealing portion [0156] **71A**: inner peripheral lip [0157] **72**: outer sealing portion [0158] **72A**: outer peripheral lip

Claims

1. A connector, comprising: a wire including a core wire and an insulation coating covering an outer periphery of the core wire; a terminal to be connected to the core wire; an inner housing

including a terminal accommodating portion for accommodating the terminal inside; a rear holder to be attached to a rear part of the inner housing; and a collective sealing portion to be mounted into the rear holder, the rear holder including a wire insertion hole, the wire being inserted through the wire insertion hole, the collective sealing portion including an inner sealing portion for sealing the wire inserted through the wire insertion hole, the insulation coating of the wire including a sheath portion and a body portion separate from each other, and the sheath portion being disposed forward of the body portion and slidable with respect to the core wire.

2. The connector of claim 1, wherein the inner sealing portion seals the body portion.

3. The connector of claim 1, further comprising: an outer housing to be assembled with the rear holder while accommodating the inner housing inside; and an outer sealing portion for sealing between the rear holder and the outer housing.

4. The connector of claim 3, wherein the outer sealing portion is integrally formed to the collective sealing portion.

5. The connector of claim 4, wherein the collective sealing portion and the rear holder are constituted by one member.

6. A connector manufacturing method for a connector with a wire including a core wire and an insulation coating covering an outer periphery of the core wire, a terminal, an inner housing including a terminal accommodating portion, a rear holder including a wire insertion hole, and a collective sealing portion to be mounted into the rear holder, the collective sealing portion sealing the wire inserted through the wire insertion hole, comprising: accommodating the terminal inside the terminal accommodating portion; attaching the rear holder having the collective sealing portion mounted therein to a rear part of the inner housing; forming a sheath portion and a body portion disposed rearward of the sheath portion by cutting off a front end part of the insulation coating of the wire; protecting a front end part of the core wire by the sheath portion by sliding the sheath portion forward with respect to the core wire; inserting the wire having the front end part of the core wire protected by the sheath portion into the wire insertion hole of the rear holder and bringing the front end part of the core wire to inside of the terminal; and connecting the core wire and the terminal.
